1a)

Number: total weight of all polymer chains divided by total # of chains

Formula: Notation:

$$\frac{\Sigma(N_i M_i)}{\Sigma N_i}$$ $\bar{M}_n$

Weight: weighing polymer chain according to its mass. Weight fraction of each component. This provides insight for the distribution of molecular weight in a sample

Formula: Notation:

$$\frac{\Sigma(N_i M_i^2)}{\Sigma(N_i M_i)}$$ $\bar{M}_w$

z-average: molecular weight determination method used in light scattering methods & it is sensitive to heavy chains.

Formula:

$$\frac{\Sigma(N_i M_i^3)}{\Sigma(N_i M_i^2)}$$ $\bar{M}_z$

3

b) polydispersity index (PDI): A measure of the distribution of the molecular weight of a polymer.

Formula: $\bar{M}_w / \bar{M}_n$ Notation: PI

2

Degree of polymerization (DP): The # of repeating units in the polymer

Formula: Notation: DP

n

2) Calculated using excel & formulas from above ↑ pt 1.

$$\sigma = \overline{M_n}\left(\frac{\overline{M_w}}{\overline{M_n}} - 1\right)^{1/2}$$

fractions	1	2	3	4	5	6	7	8	9
mass (g), wi	1.082	0.904	0.801	0.31	0.041	1.043	1.281	1.171	0.241
Mi x10^-4 (g/mol)	1.042	2.312	2.619	3.283	3.501	3.984	5.017	7.984	9.308

Mn	2.71586	x 10^4 (g/mol)
Mw	4.16805	x 10^4 (g/mol)
Mz	5.5728	x 10^4 (g/mol)
PDI	1.53471	
σ	1.98594	x 10^4 (g/mol)

10

3)

a) $$\frac{1}{M_n} = \frac{W_A^*}{M_{n,A}} + \frac{W_B^*}{M_{n,B}} = \frac{W_A^*}{M_{n,A}} + \frac{1-W_A^*}{M_{n,B}} = \frac{1}{178{,}000} = \frac{W_A}{128{,}000} + \frac{1-W_A}{225{,}000}$$

solving for W_A algebraically $W_A = 0.3484$

$W_B = 1 - 0.3484 = 0.6516$

5

b) $$PDI = \frac{\overline{M_w}}{\overline{M_n}} = \frac{\overline{M_w}}{W_A \overline{M_{n,A}} + W_B \overline{M_{n,B}}} = \frac{334\,603.93}{(0.3484)(128000) + (0.6516)(225000)}$$

$PDI = 1.76$

4)

$$M_{n,mix} = \frac{1}{W_A / M_{n,A} + W_B / M_{n,B}}$$

$$M_{w,mix} = W_A M_{w,A} + W_B M_{w,B}$$

$$PDI = \frac{M_w}{M_n}$$

Total weight

30+25 = 55g

$W_A = \frac{25}{55} = 0.45$ $W_B = \frac{30}{55} = 0.54$

$$M_{n,mix} = \frac{1}{\frac{0.45}{65670} + \frac{0.54}{45000}} = \frac{1}{1.904 \cdot 10^{-5}} = 52512.2 \text{ g/mol}$$

$M_{w,mix} = (0.45)(75000) + (0.54)(88000) = 82.091\,5 \text{ g/mol}$ 5

$$PDI = \frac{82.0915 \text{ g/mol}}{52512.2 \text{ g/mol}} = 1.56$$

5) $M = M_A + M_B + M_C = 1342 + 3245 + 2458$

$= 7045$

$$z_A = \frac{1342}{7045} = 0.1905 \quad z_B = \frac{3245}{7045} = 0.46 \quad z_C = \frac{2458}{7045} = 0.3489$$

$$\bar{M}_n = \frac{1}{\Sigma(w_i/M_i)} = 49112.47 \text{ g/mol}$$ 5

$$\bar{M}_w = \Sigma(w_i M_i) = 64925.69 \text{ g/mol}$$

6) $$M_w = \frac{\Sigma_i N_i M_i^2}{\Sigma_i N_i M_i} \rightarrow M_{w,mix} = \frac{\Sigma_{iA}(N_i M_i^2)^A + \Sigma_{jB}(N_j M_j^2)^B}{\Sigma_{iA}(N_i M_i)^A + \Sigma_{jB}(N_j M_j)^B}$$

$$W_A = \Sigma_{iA}(N_i M_i)^A, \quad W_B = \Sigma_{jB}(N_j M_j)^B$$

$$M_{wA} = \frac{\Sigma_{iA}(N_i M_i^2)^A}{\Sigma_{iA}(N_i M_i)^A} = \frac{\Sigma_{iA}(N_i M_i^2)^A}{W_A} \rightarrow \begin{array}{l} M_{w,A} W_A = \Sigma_{iA}(N_i M_i^2)^A \\ M_{wB} W_B = \Sigma_{jB}(N_j M_j^2)^B \end{array}$$

plug in

$$M_{w,mix} = \frac{M_{w,A} W_A + M_{w,B} W_B}{W_A + W_B}$$

$$W_A^* = \frac{W_A}{W_A + W_B} \rightarrow W_B^* = \frac{W_B}{W_A + W_B}$$

$\downarrow$

$$M_{w,mix} = W_A^* M_{w,A} + W_B^* M_{w,B}$$ 3

gen formula: $M_{w,mix} = \Sigma_i W_i^* M_{w,i}$